

concurrency and parallelisation.

The latest stable release of Clojure is 1.8.

The syntax for the ‘Hello World’ program in Clojure is:

```
On repl
Start repl. When you use Leiningen, type lein repl on
Terminal (Command Prompt):

bash-3.2$ lein repl
nREPL server started on port 59553 on host 127.0.0.1
REPL-y 0.2.1
Clojure 1.5.1
Docs: (doc function-name-here)
      (find-doc "part-of-name-here")
Source: (source function-name-here)
Javadoc: (javadoc java-object-or-class-here)
Exit: Control+D or (exit) or (quit)
Results: Stored in vars *1, *2, *3, an exception in *e
user=>
Type below on repl:
user=> (prn "Hello World")
"Hello World"
nil
user=> (println "Hello World")
Hello World
nil
user=> (pr-str "Hello World")
""
"\Hello World""
```

The advantages of Clojure

- Clojure code can be used on any Java library. Clojure libraries can, in turn, be used from Java and Clojure applications can be packed like Java applications and deployed anywhere.
- Clojure is part of the Lisp environment and it retains the best features of Lisp. Clojure contains macros, which is regarded as an important approach to meta programming and syntactic extensions.
- Clojure, being a functional programming language, enables developers to use first-class and high order functions with values, and comes with its own set of efficient immutable data structures.
- Being a dynamic programming language, it supports updating and loading of new code at runtime, either locally or remotely.

Clojure's disadvantages

- Debugging in Clojure with regard to error handling and removal is a tedious task.
- Clojure can only run on the Java virtual machine.
- Clojure is hard to master for newbies, and sometimes even developers take a lot of time when using it for core and professional program development.

8. Haskell

Haskell is a modern, standard, non-strict, purely functional programming language, and is built with features like polymorphic typing, lazy evaluation and higher order functions. Haskell features a type system with type inference and lazy evaluation, and also includes type classes. It is purely based on semantics, but not the syntax of the language of Miranda which focused on the efforts of the initial Haskell working group.

Haskell has a strong, static type system based on the Hindley–Milner type inference. Haskell's principal innovation in this area is to add type classes, originally conceived as a principled way to add overloading to the language, but since then this function has been found to address many more uses.

Haskell has an open, published specification, and multiple implementations exist. Its main implementation, the Glasgow Haskell Compiler (GHC), is both an interpreter and native-code compiler that runs on most platforms. GHC is noted for its high-performance implementation of concurrency and parallelism, and for having a rich type system incorporating recent innovations such as generalised algebraic data types and type families. Another interesting feature of Haskell is that all functions are treated as values like integers and strings.

The latest version is Haskell 2014.

The syntax for the ‘Hello World’ program is:

```
module Main where
main :: IO ()
main = putStrLn "Hello, World!"
```

The advantages of Haskell

- Being a purely functional programming language, Haskell enables users to get things done by giving the computer a sequence of tasks and then executing them. While executing, the tasks change state.
- Haskell is a lazy programming language and will not execute functions and calculate things until really forced to show the results. It goes well with referential transparency, and allows developers to think of programs as a series of transformations on data.
- Haskell is elegant and concise. Its programs are shorter compared to other programming languages and debugging becomes a lot easier.

Haskell's disadvantages

- Haskell is not suitable for making time-critical applications.
- Being a new programming language, it is very difficult to understand, compared to other languages like C or C++, and has limited community support and documentation.
- There are very few platforms where Haskell code has been used.